

Spatial Interpolations Techniques

INTERPOLATION

- In many instances we may wish to model a variable as a continuous field (i.e. a 'surface'), yet we only have data values for a finite number of points. It therefore becomes necessary to **interpolate** (i.e. estimate) the values for the intervening points.
- Interpolation is the process of obtaining a value for a variable of interest at a location where data has not been observed, using data from locations where data has been collected.
- Usual Rationale: points close together are more likely to have similar values than points far apart (**Tobler's Law**)

General Classification of Interpolation Techniques

- Point/Line/Area
- Global / Local
- Exact / Approximate
- Deterministic / Stochastic
- Gradual / Abrupt

Point Based Interpolation

Given points whose locations and values are known, determine the values of other points at locations

For example:

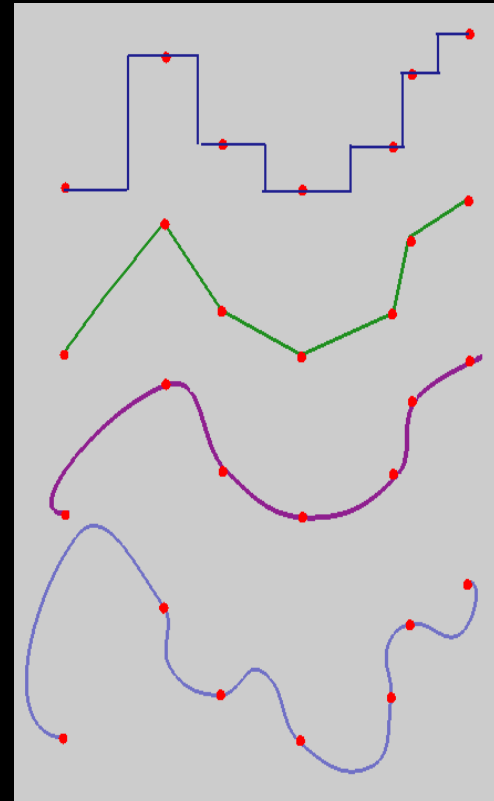
- weather station readings
- spot heights
- porosity measurements

Line Based Interpolation

The input data does not necessarily need to be points. In fact, points, lines and/or areas can be interpolated.

Paper maps often have the elevation represented as contour lines, where the elevation is the same along a particular line.

If such contour lines were digitized, they could be interpolated to create a continuous surface (DEM). But often, contour lines/isolines are the result of a first interpolation!

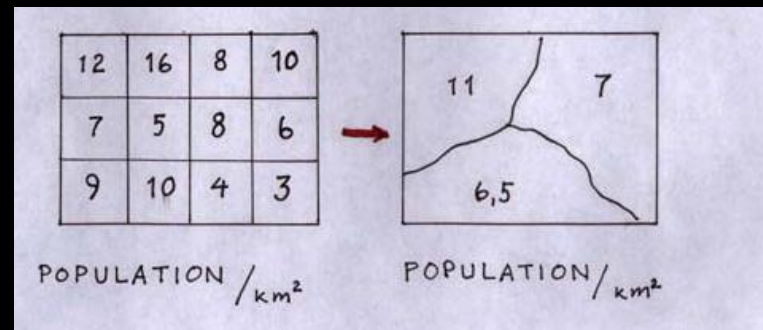


Area Based Interpolation

Given a set of data mapped on one set of source zones, determine the values for a different set of target zones.

For example:

- given population counts for census tracts, estimate populations for electoral districts



Area Based Interpolation

How it works?

A two step process

Identifying Centroid:

1. find centroid of area
2. assign total value of data in area to centroid
3. treat as point interpolation.

Overlay:

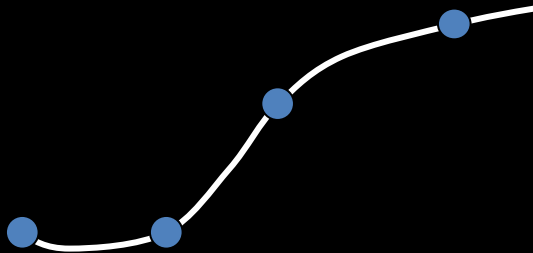
1. overlay of target and source zones
2. determine the proportion of each source zone that is assigned to each target zone
3. apportion the total value of the attribute for each source zone to target zones

Global vs. Local Interpolators

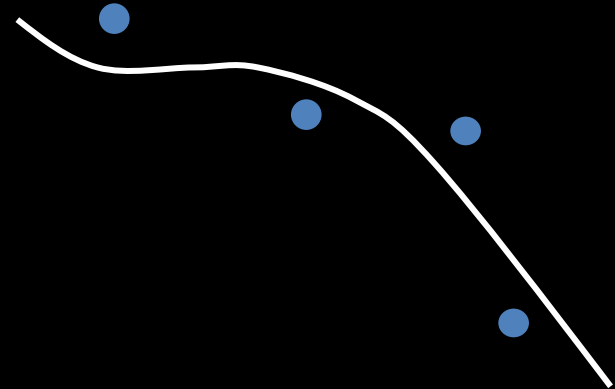
- Global interpolators determine a single function which is mapped across the whole region
 - e.g. trend surface
- Local interpolators apply an algorithm repeatedly to a small portion of the total set of points
 - e.g. inverse distance weighted

Exact vs. Inexact

Exact interpolation predicts a value at the point location that is the same as its known value. In other words, exact interpolation generates a surface that passes through the control points.



Inexact interpolation or approximate interpolation predicts a value at the point location that differs from its known value.



Deterministic vs. Stochastic

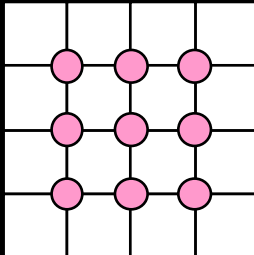
- **Deterministic interpolators** model a data point at a particular position.
 - e.g. spline, theissen polygons
- **Stochastic interpolators** try to model probability of a data point being at a particular position.
- Stochastic methods incorporate the concept of randomness.
- Use both statistical and mathematical methods that can be used to create surfaces and assess the uncertainty of the predictions.
 - e.g. kriging methods

Gradual/Abrupt Interpolators

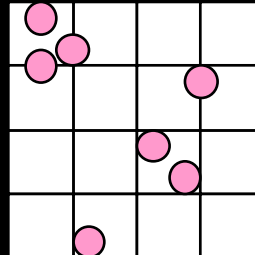
- Gradual interpolators assume continuous and smooth behavior of data everywhere.
- Abrupt interpolators allow for sudden changes in data due to boundaries or undefined derivatives.

Data sampling

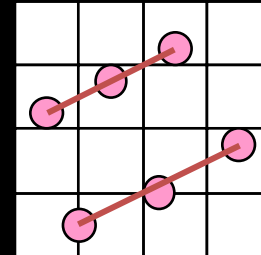
- Method of sampling is critical for subsequent interpolation...



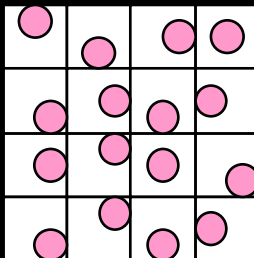
Regular



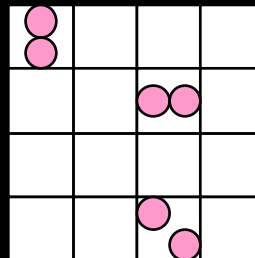
Random



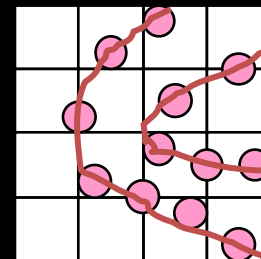
Transect



Stratified random



Cluster



Contour

Example Interpolators

- Theissen Polygons
- Triangulated Irregular Networks
- Inverse Distance Weighted
- Splines
- Radial Basis Functions
- Global Polynomial
- Kriging

Thiessen Polygons

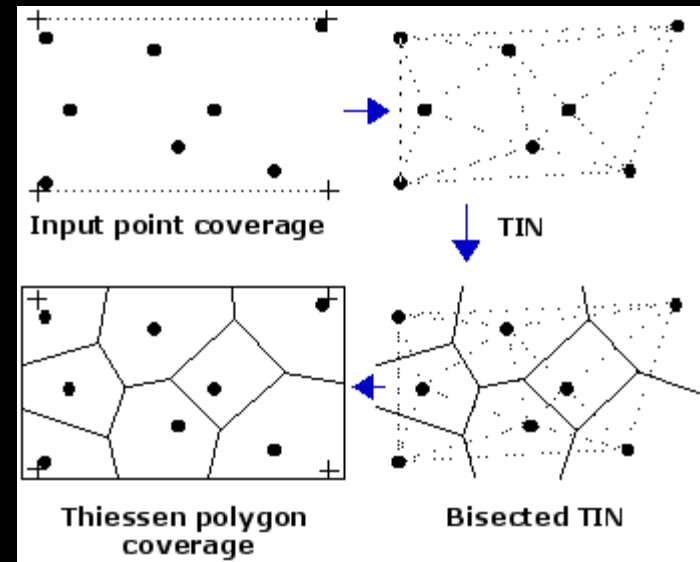
- Thiessen (Voronoi) polygons:
 - assume values of un-sampled locations are equal to the value of the nearest sampled point.
- Vector-based method
 - regularly spaced points produces a regular mesh.
 - irregularly spaced points produces an network of irregular polygons.

Thiessen Polygons

- Local, exact, deterministic.
- Thiessen polygons (Dirichlet or Voronoi diagrams) are constructed around the known data points.
- All points within the same polygon are closer to that data point than to any other and are assumed to have the same value.

Construction of Thiessen polygons

- All points are triangulated into a triangulated irregular network (TIN) that meets the Delaunay criterion.
- The perpendicular bisectors for each triangle edge are generated, forming the edges of the Thiessen polygons. The locations at which the bisectors intersect determine the locations of the Thiessen polygon vertices.



Delaunay criterion:

- Intuitively: given a set V of points, among all the triangulations that can be generated with the points of V , the Delaunay triangulation is the one in which triangles are as much equiangular as possible
- In other words, Delaunay triangulations tend to avoid long and thin triangles: important for numerical problems

Inverse Distance Weight (IDW) Method

– A Deterministic Model

- Explicitly implements the assumption that things that are close to one another are more alike than those that are farther apart.
- To estimate value at un-measured location, IDW uses the values of measured values surrounding the prediction location.
- The measured values closest to the prediction location have more influence (greater weight) on the predicted value than those farther away.
- IDW assumes that each measured point has a local influence that diminishes with distance. It gives greater weights to points closest to the prediction location, and the weights diminish as a function of distance, hence the name **inverse distance weighted**.
- Weights are proportional to the inverse of the distance (between the data point and the prediction location) raised to the power value p
- IDW is an exact interpolator, where the maximum and minimum values in the interpolated surface can only occur at sample points.

IDW Equation

$$z = [w_1 * z(1) + \dots + w_n * z(n)] / [w_1 + \dots + w_n]$$

$W_1, W_2, \dots, W_n$ are the weights of sample located at 1, 2, 3, ..., n.

$z_1, z_2, z_3, \dots, z_n$ are the values of the quantity at locations 1, 2, 3, ..., n

$$W_i = 1/d_i^p$$

Weights are inversely proportional to the power 'p' of the distances between the point of interpolation and the data locations

Inverse Distance Weight (IDW) Method

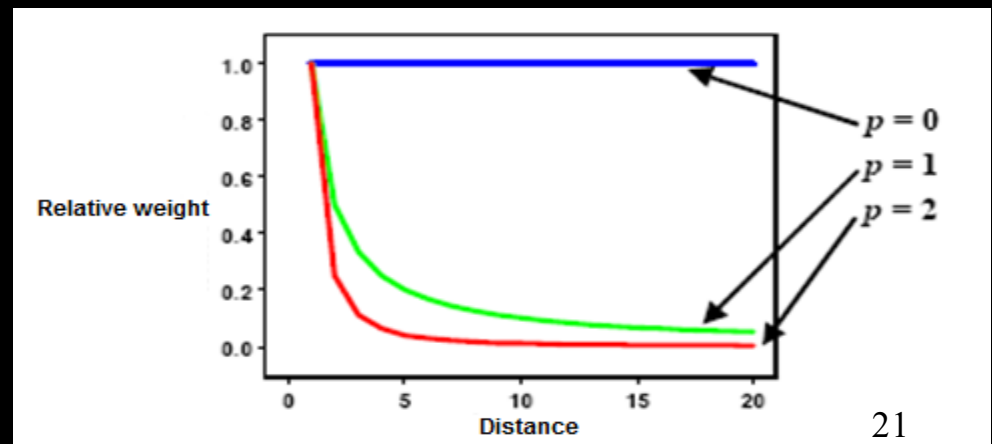
The Power Function - How it works?

Weights are proportional to the inverse of the distance (between the data point and the prediction location) raised to the power value p . As a result, as the distance increases, the weights decrease rapidly. The rate at which the weights decrease is dependent on the value of p .

If $p = 0$, there is no decrease with distance, and because each weight λ_i is the same, the prediction will be the mean of all the data values in the search neighborhood.

As p increases, the weights for distant points decrease rapidly. If the p value is very high, only the immediate surrounding points will influence the prediction.

An optimal power value can be determined by minimizing the root mean square prediction error (RMSPE). The RMSPE is a statistic that is calculated during cross-validation.



Inverse Distance Weight (IDW) Method

Search Neighbourhood- What it is?

- To speed calculations, you can exclude the more distant points that will have little influence on the prediction. As a result, it is common practice to limit the number of measured values by specifying a search neighborhood.
- The shape of the neighborhood restricts how far and where to look for the measured values to be used in the prediction.
- The shape of the neighborhood is influenced by the input data and the surface you are trying to create. If there are no directional influences in your data, you'll want to consider points equally in all directions.
- If no directional influence- Search neighborhood shape will be a circle.
- However, if there is a directional influence in your data, the shape of the search neighborhood shall be an ellipse with the major axis parallel to the phenomena influence the data. The adjustment for this directional influence is justified because you know that locations parallel from a prediction location are going to be more similar at remote distances than locations that are perpendicular to the influencing phenomena but located closer to the prediction location.

Inverse Distance Weight (IDW) Method

Example

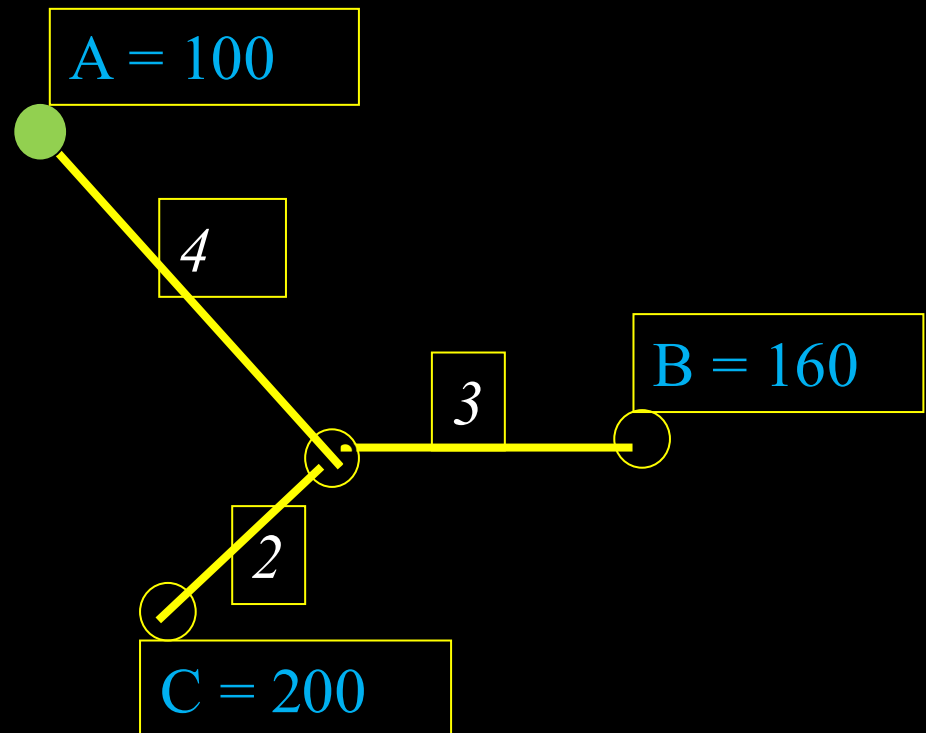
Weights

A $1 / (4^2) = .0625$

B $1 / (3^2) = .1111$

C $1 / (2^2) = .2500$

Unknown Value = ?



Inverse Distance Weight (IDW) Method - Example

Contd.

	Weights	Weights * Value
A	$1 / (4^2) = .0625$	$.0625 * 100 = 6.25$
B	$1 / (3^2) = .1111$	$.1111 * 160 = 17.76$
C	$1 / (2^2) = .2500$	$.2500 * 200 = 50.00$

$$\text{Total} = .4236$$

$$6.25 + 17.76 + 50.00 = 74.01$$

Unknown Value

$$74.01 / .4236 = \mathbf{175}$$

Drawbacks of Deterministic Interpolators

None of the deterministic methods provide sensible information on:

- The number of points needed to compute the local average,
- The size, orientation and shape of the neighbourhood from which those points are drawn,
- Whether there are better ways to estimate the interpolation weights other than as a simple function of distance, and
- The errors (uncertainties) associated with the interpolated values

How to overcome this?

**The solution is through Geostatistics based
Interpolation**

What is Geostatistics?

- Geostatistics “study of phenomena that vary in space and/or time”. – Deutsch, (2002).
- Geostatistics is the collection of techniques for the solution of estimation problems involving spatial variables. (Olea, 1999)
- “Geostatistics offers a way of describing the spatial continuity of natural phenomena and provides adaptations of classical regression techniques to take advantage of this continuity.” (Isaaks and Srivastava, 1989).

POP QuiZ

1. What is buffering in GIS analysis?
2. What is overlay analysis in GIS?
3. List three error in digitization?
4. What are sliver polygons?